

THE UNIVERSITY OF
MELBOURNE

Final Report

The Art of Scientific Computing: Project Title: Opinion Dynamics

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0.1 The Voter Model

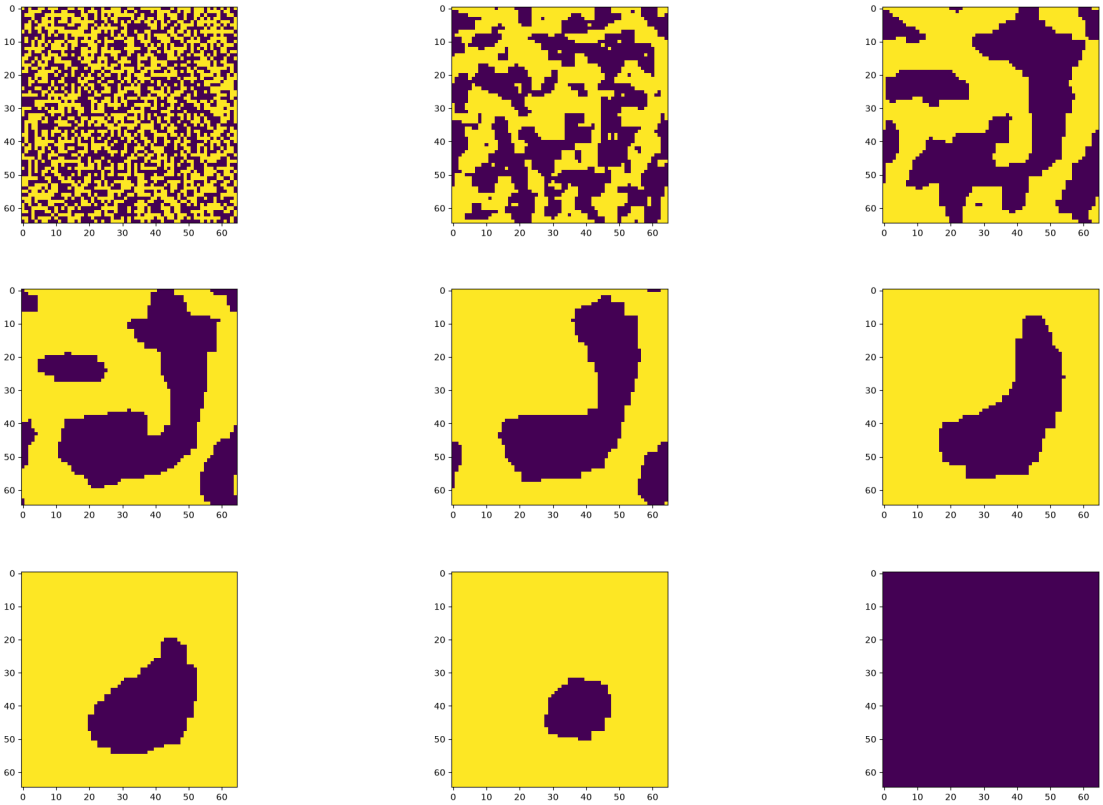
Our first and simpilist model of opinion dynamics is the well studied voter model INCLUDE REFERENCES

0.1.1 Definitions

Define the Voter Model

0.1.2 Limiting Behaviour

Reference resuluts from the literature on theorems regarding limiting behaviour Show plots reproducing said behaviour



0.2 CODA Model

0.2.1 definition

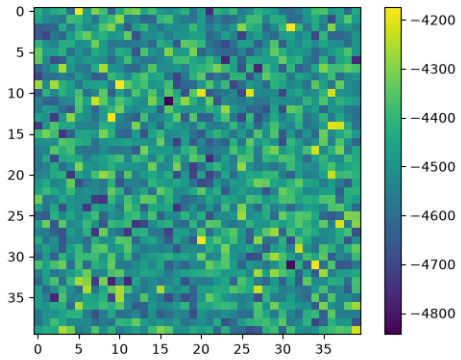
Define the basic coda model

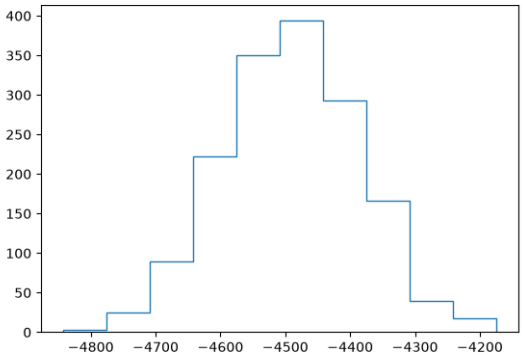
0.2.2 phase transition

Define average distance to disagreement statistic and argue why we expect this to give a phase transition as parameters vary.

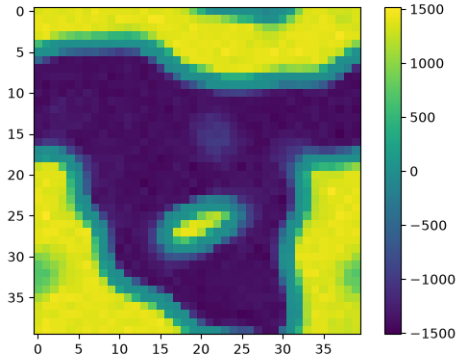
0.2.3 variant model

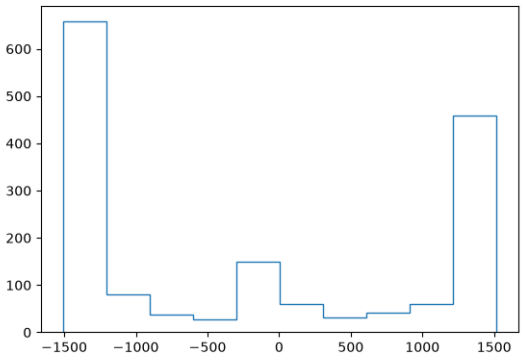
define variant model with change based on opinions, discuss how this reduces to the voter model in the limit



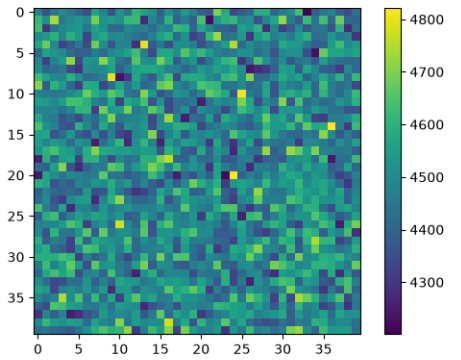


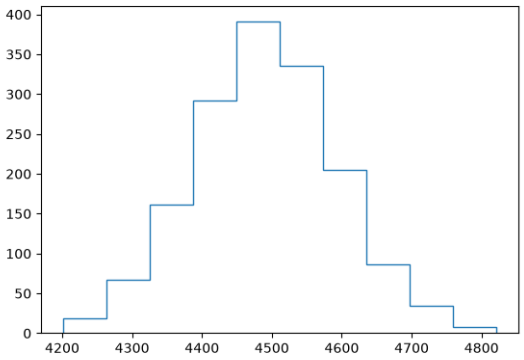
(a) Favouring negative



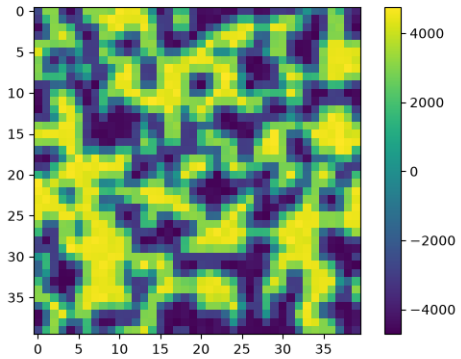


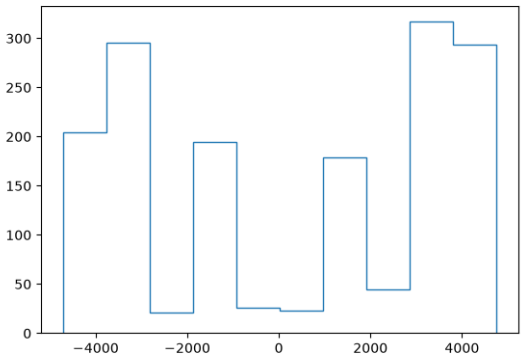
(b) Favouring disagreement





(c) Favouring positive





(d) Favouring agreement